



Fig. 8: Image

function can be used for histogram shrinking.

Histogram sliding. This technique consists of simply adding or subtracting a constant brightness value to all pixels in the image. The overall effect is an image with comparable contrast properties, but higher or lower average brightness, respectively. `imadd` and `imsubtract` functions can be used for histogram sliding.

When implementing histogram sliding, you must make sure that pixel values do not go outside the greyscale boundaries. An example of histogram sliding is given below:

```
A = imread('Penguins_grey.jpg');
imshow(A), title('Original Image');
B=im2double(A);
bright_add = 0.2;
imhist(A), title('Original Histogram');
C=B+bright_add;
imshow(C), title('New Bright Image');
imhist(C), title('New Histogram');
```

In the example, the image is brightened by adding 0.2 to its pixel values. Figs 10 and 11 show the original image and its histogram, respectively. Figs 12 and 13 show the modified image and its histogram, respectively.

Thresholding

Thresholding is used to remove unnecessary details from an image and concentrate on essentials. It is also used to bring out hidden details, in case the object of interest and

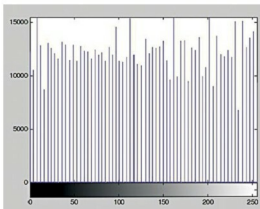


Fig. 9: Equalised histogram



Fig. 10: Original image

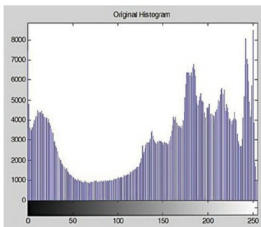


Fig. 11: Histogram of original image shown in Fig. 10

background have similar grey levels. Thresholding can be further classified as single thresholding and double thresholding. In MATLAB, single as well as double image thresholding can be done.

Single thresholding. A greyscale image is turned into a binary image

(black and white) by first choosing a grey level 'T' in the original image, and then turning every pixel black or white depending on whether its grey value is greater than or less than 'T'. Thresholding is a vital part of image segmentation, where users wish to isolate objects from the background. To convert an image F into black-and-white image G with threshold of 100, the command in MAT-

LAB is `G = F > 100`.

The following example reads image `Penguins_grey.jpg` and displays both the original image and the image generated after thresholding using a factor of 70:

```
>> A = imread('Penguins_grey.jpg');
>> imshow(A), figure, imshow(A>70)
```

Figs 14 and 15 show the original image and the image after thresholding, respectively.

In addition, there is a command in MATLAB which converts a greyscale image or a coloured image into black-and-white image. The command is `im2bw(image,level)`, where `image` is the greyscale image and `level` is a value between 0 and 1.

Double thresholding.

In this case, there are two values `T1` and `T2`, and the thresholding operation is performed as the pixel becomes white if the grey level is between `T1` and `T2`. Also, the pixel becomes black if the grey level is outside these threshold values.

Image sharpening

Image sharpening is a powerful tool for emphasising texture and drawing viewer focus. It can improve image quality, even more than what is achieved through upgrading to a high-end camera lens.